UNIVERSITY OF PASSAU
Faculty of Computer Science and Mathematics

Energy Informatics I WS2026/27





Hermann de Meer, Michael Bettermann, Stepan Gagin

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Course Details (I)

Contact

- Lecturer
  - Hermann de Meer
    - Office: ITZ 157
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- Tutors
  - Michael Bettermann
    - Office: ITZ 139b
    - Email: Michael.Bettermann@uni-passau.de
  - Stepan Gagin
    - Office: ITZ 150
    - Email: Stepan.Gagin@uni-passau.de

Add the following tag in front of the subject of all your emails **[EnInf I - WS26/27]**

Course Details (II)

Course Format

- Language: **English**
- Lecture:
  - Wednesday 10:00 – 12:00, weekly
- Tutorial:
  - Tuesday 14:00 – 16:00
  - Tuesday 16:00 – 18:00
  - Thursday 16:00 – 18:00
- Credits: **6 ECTS**
- Communication: Slides, exercises, groups, and all up-to-date information will be made available via **Stud.IP**
- Written exam (90 minutes)
  - 17.02.2027, 10:00 to 13:00, (PHIL) HS 1
  - 24.03.2027, 10:00 to 13:00, (PHIL) HS 1

Course Details (III)

Course Objectives

- This course can be seen as a complementary of **FoES** (Foundation of Energy Systems) (now **Basic Principles of Power System Operation**):
  - FoES presents the **basics** of the **electrical grid**: physical phenomena, how energy flows, the different components of a grid, etc.
  - The present course is about presenting how to make this **electrical systems smart** using **ICT**, thus the title **Energy informatics**
- Learn the **architectures, concepts**, and **protocols** for designing and efficiently managing an electrical grid
- A special attention will be dedicated to computer networks and telecommunication technologies, and smart grid modelling techniques
- Some topics are covered as an excursus, which are interesting from a research point of view, but for various reasons are of low practical importance.

Content covered

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- Motivation
- Distributed energy resources
- Flexibility
- Energy economics
- Smart grid architectural model
- Standards and technology
- Advanced metering infrastructure
- Electric vehicle charging
- Substation automation
- Smart converters